

Study Overview

Brief Summary

This is a Phase 1a/1b study of aurora kinase A inhibitor VIC-1911 administered as monotherapy and in combination with sotorasib for the treatment of locally advanced or metastatic KRAS G12C-mutant non-small cell lung cancer (NSCLC).

Detailed Description

Selected subjects will include males and females age ≥ 18 years; histologically confirmed locally advanced or metastatic KRAS G12C-mutated NSCLC; received at least 1 prior line of cancer therapy with a PD-1 or PD-L1 inhibitor with or without platinum-based chemotherapy; recovered from all acute toxicities ($\leq$ Grade 1) due to prior therapy; adequate renal and hepatic function; and no known history of significant cardiac, hepatic or ocular disease.

Dose Escalation Phase:

Following screening, a total of up to 36 subjects are anticipated to establish the dose limiting toxicity (DLT) and maximum tolerated doses (MTDs) of VIC-1911 monotherapy and VIC-1911 in combination with sotorasib therapy.

Cohort 1a: Subjects who are refractory to or relapsed on prior KRAS G12C inhibitor therapy will receive VIC-1911 monotherapy. Up to 24 subjects are anticipated in this cohort.

Cohort 1b: Subjects who are refractory to or relapsed on prior KRAS G12C inhibitor therapy or are naïve to KRAS G12C inhibitor therapy will receive VIC-1911 plus sotorasib combination therapy. Up to 12 subjects are anticipated in this cohort.

A 3+3 dose escalation schema will be followed to establish the dose limiting toxicity (DLT) and maximum tolerated dose (MTD) of VIC-1911 and VIC-1911 plus sotorasib combination. A total of at least 6 subjects will be treated at the MTD in each group before initiating the Expansion Phase.

Expansion Phase:

Following screening, a total of 104 subjects with KRAS G12C-mutated locally advanced or metastatic NSCLC are anticipated to expand the disease treatment settings of VIC-1911 as monotherapy or in combination with sotorasib. VIC-1911 monotherapy and VIC-1911 plus sotorasib combination therapy will be administered orally at the MTDs established during the Dose Escalation Phase.

Cohort 2a: Subjects who are refractory to or relapsed on prior KRAS G12C inhibitor therapy will receive VIC-1911 monotherapy. (n=29)

Cohort 2b: Subjects who are refractory to or relapsed on prior KRAS G12C inhibitor therapy will receive VIC-1911 plus sotorasib combination therapy. (n=29)

Cohort 2c: Subjects who are naïve to KRAS G12C inhibitor therapy will receive VIC-1911 plus sotorasib combination therapy. (n=46)

VIC-1911 and sotorasib will be taken in the fasted state, 1 hour before or 2 hours after a meal.

Subjects who demonstrate clinical benefit (CR, PR or SD) will be allowed to continue therapy with VIC-1911 and sotorasib until progression of disease, observation of unacceptable adverse events, intercurrent illness or changes in the subject's condition that prevents further study participation.

Disease response will be assessed according to Response Evaluation Criteria in Solid Tumors (RECIST v.1.1).

Blood for hematology, coagulation parameters and serum chemistry determinations and urine will be collected, ECGs will be taken and ophthalmologic exams will be

conducted during the study.

Blood will be taken for PK assessment of VIC-1911 and PD assessment of circulating tumor DNA biomarker determinations.

Tumor biopsies will be taken from consenting subjects at Screening and on-study for correlative biomarker determinations. Results will be correlated with clinical outcome.

Official Title

A Phase 1a/1b Study of Aurora Kinase A Inhibitor VIC-1911 Monotherapy and in Combination With Sotorasib for the Treatment of KRAS G12C-Mutant Non-Small Cell Lung Cancer

Contacts and Locations

This section provides contact details for people who can answer questions about joining this study, and information on where this study is taking place.

To learn more, please see the Contacts and Locations section in How to Read a Study Record.

This study has 5 locations

United States

California Locations

Sacramento, California, United States, 95817

University of California Davis

Connecticut Locations

North Haven, Connecticut, United States, 06473

Yale Cancer Center

Georgia Locations

Atlanta, Georgia, United States, 30322

Emory University Winship Cancer Center

Maryland Locations

Baltimore, Maryland, United States, 21201

University of Maryland Cancer Center

New York Locations

New York, New York, United States, 10016

New York University Langone Health Perlmutter Cancer Center

Participation Criteria

Researchers look for people who fit a certain description, called eligibility criteria. Some examples of these criteria are a person's general health condition or prior treatments.

For general information about clinical research, read Learn About Studies.

Eligibility Criteria

Description

Inclusion Criteria:

Males and females ≥ 18 years of age

Have locally advanced or metastatic histologically or cytologically confirmed NSCLC, KRAS G12C-mutated

The presence of a KRAS G12C mutation should be established prior to entry as assessed in a CLIA qualified laboratory. Testing may be done on tumor tissue (archival or fresh) or on ctDNA from blood.

Have received at least 1 prior line of cancer therapy with a PD-1 or PD-L1 inhibitor with or without platinum-based chemotherapy (unless subject is not eligible or refuses chemotherapy or PD-1/PD-L1 therapy and have documented progression on all prior cancer therapies

Dose Escalation Phase:

Cohort 1a: (VIC-1911 monotherapy): Locally advanced or metastatic NSCLC refractory to or relapsed on at least 1 prior cancer therapy as noted above, and relapsed/refractory on KRAS G12C inhibitor therapy as the most recent cancer

therapy prior to study

Cohort 1b: (VIC-1911 plus sotorasib): Locally advanced or metastatic NSCLC:

Refractory to or relapsed on at least 1 prior cancer therapy as noted above, and relapsed/refractory on KRAS G12C inhibitor therapy as the most recent cancer therapy prior to study, or

Refractory to or relapsed on at least 1 prior cancer therapy as noted above, and Naïve to KRAS G12C inhibitor therapy

Expansion Phase 1b:

Cohort 2a: (VIC-1911 monotherapy): Locally advanced or metastatic NSCLC refractory to or relapsed on at least 1 prior cancer therapy as noted above, and relapsed/refractory on KRAS G12C inhibitor therapy as the most recent cancer therapy prior to study

Cohort 2b: (VIC-1911 plus sotorasib): Locally advanced or metastatic NSCLC refractory to or relapsed on at least 1 prior cancer therapy as noted above, and relapsed/refractory on KRAS G12C inhibitor therapy as the most recent cancer therapy prior to study

Cohort 2c: (VIC-1911 plus sotorasib): Locally advanced or metastatic NSCLC refractory to or relapsed on at least 1 prior cancer therapy as noted above, and naïve to KRAS G12C inhibitor therapy

Measurable disease by Response Evaluation Criteria in Solid Tumors (RECIST) 1.1

Have discontinued previous treatments for cancer, except for sotorasib for subjects to receive VIC-1911 plus sotorasib combination treatment, and have resolution, except where otherwise stated in the inclusion criteria, of all clinically significant toxic effects of prior cancer treatment, surgery, or radiotherapy to Grade ≤ 1

Adequate performance status: Eastern Cooperative Oncology Group (ECOG) ≤ 2

Life expectancy of ≥ 3 months

Subjects with brain metastases:

11.1 KRAS G12C inhibitor naïve: Subjects with clinically stable (i.e., no increase in corticosteroid requirement) asymptomatic brain metastases are allowed without prior local therapy, as long as all lesions are each ≤ 1 cm. Prior local therapy is required (e.g., stereotactic radiosurgery [SRS], stereotactic body radiation therapy [SBRT], or surgery) for any lesion > 1 cm or any lesion that is symptomatic. Subjects must be clinically stable and asymptomatic following local therapy.

11.2 KRAS G12C inhibitor pretreated: Subjects with clinically stable (i.e., no increase in corticosteroid requirement) asymptomatic brain metastases following prior local therapy (e.g., SRS, SBRT or surgery) are allowed.

Adequate hematologic without ongoing transfusion support:

Hemoglobin (Hb) ≥ 8 g/dL

Absolute neutrophil count (ANC) $\geq 1.0 \times 10^9$ cells/L

Platelets $\geq 75 \times 10^9$ cells/L

Adequate renal and hepatic function:

Calculated creatinine clearance ≥ 50 mL/minute $\times 1.73$ m² per the Cockcroft-Gault formula

Total bilirubin ≤ 1.5 times the ULN, unless due to Gilbert's disease, or < 3 times the ULN for subjects with liver metastases

ALT/AST ≤ 2 times the ULN, or < 3 times the ULN for subjects with liver metastases

Negative serum pregnancy test within 14 days prior to the first dose of study therapy for women of child-bearing potential (WCBP). Sexually active WCBP and male subjects must agree to use adequate methods to avoid pregnancy throughout the study and for 28 days after the completion of study treatment.

Ability to provide written informed consent

Exclusion Criteria:

Serious cardiac condition within the last 6 months, such as uncontrolled

arrhythmia, myocardial infarction, unstable angina or heart disease defined by the New York Heart Association (NYHA) Class III or Class IV
 QT interval corrected for rate (QTc) > 480 msec on the ECG obtained at Screening using Fridericia method for QTc calculation
 Medications that are inhibitors or inducers of UDP-glucuronosyltransferases (UGTs) are prohibited in the Dose Escalation Phase
 History of corneal epithelial cysts or other ocular events leading to blurred vision, or has medically relevant abnormalities identified on screening ophthalmologic examination
 Symptomatic pneumonitis/interstitial lung disease requiring medical intervention
 Symptomatic central nervous system metastasis
 Leptomeningeal carcinomatosis
 Inability to swallow oral medication
 Gastrointestinal conditions that could impair absorption or tolerance of study drugs
 Current hematologic malignancies
 Second, active primary solid tumor malignancy that, in the judgement of the Investigator or Sponsor Medical Monitor, may affect the interpretation of results, with the exception of carcinoma in situ of any origin, non-muscle invasive bladder cancer, and Gleason < 3+3 prostate cancer
 Active infection with human immunodeficiency virus (HIV), hepatitis B virus (HBV) or hepatitis C virus (HCV) requiring treatment within the last week prior to study treatment
 Other active infection requiring IV antibiotic usage within the last week prior to study treatment
 Unable to tolerate marketed dose of KRAS G12C inhibitor on prior therapy for subjects to be enrolled in combination VIC-1911 plus sotorasib treatment cohorts. Alternatively, these subjects may be able to enroll in the VIC-1911 monotherapy treatment cohort, upon discussion with the Medical Monitor and Study Chair.
 Previous MEK or EGFR inhibitor therapy
 Any other medical intervention or other condition which, in the opinion of the Principal Investigator, could compromise adherence to study requirements or confound the interpretation of study results
 Receipt of an investigational product on a clinical trial within 5 elimination half-lives or within 28 days, whichever is shorter, prior to C1D1 on this study, or currently enrolled in a clinical trial involving an investigational product or any other type of medical research judged not to be scientifically or medically compatible with this study
 Previously completed or withdrawn from any other study investigating an aurora kinase A inhibitor
 Known hypersensitivity to VIC-1911 or its components
 If female, pregnant, breast-feeding, or planning to become pregnant

Study Plan

This section provides details of the study plan, including how the study is designed and what the study is measuring.

Expand all / Collapse all

How is the study designed?

Design Details

Primary Purpose : Treatment

Allocation : Non-Randomized

Interventional Model : Single Group Assignment

Interventional Model Description: Dose Escalation Phase followed by Expansion Phase. A total of 24 subjects are anticipated in Dose Escalation Phase, Cohort 1a. A total of 12 subjects are anticipated in Dose Escalation Phase, Cohort 1b. A total of 29 subjects are anticipated in Expansion Phase, Cohort 2a. A total of 29 subjects are anticipated in Expansion Phase, Cohort 2b. A total of 46 subjects are anticipated in Expansion Phase, Cohort 2c.

Masking : None (Open Label)

Arms and Interventions

Participant Group/Arm Intervention/Treatment

Experimental: Dose Escalation Phase, Cohort 1a: VIC-1911 monotherapy
Subjects with locally advanced or metastatic KRAS G12C-mutated NSCLC refractory to or relapsed on prior KRASG12C inhibitor therapy will receive VIC-1911 monotherapy.
Drug: VIC-1911
VIC-1911 tablets for oral administration

Experimental: Dose Escalation Phase, Cohort 1b: VIC-1911 plus sotorasib combination therapy
Subjects with locally advanced or metastatic KRAS G12C-mutated NSCLC refractory to or relapsed on prior KRASG12C inhibitor therapy or are naive to KRAS G12C inhibitor therapy will receive VIC-1911 plus sotorasib combination therapy.

Only Dose Level 1 was opened and all 3 subjects received sotorasib 960 mg QD PO, VIC-1911 75 mg BID PO.

Drug: VIC-1911
VIC-1911 tablets for oral administration

Drug: sotorasib
Sotorasib tablets for oral administration

Other Names:

LUMAKRAS

Experimental: Expansion Phase, Cohort 2a: VIC-1911 monotherapy
Subjects with locally advanced or metastatic KRAS G12C-mutated NSCLC refractory to or relapsed on prior KRASG12C inhibitor therapy will receive VIC-1911 monotherapy.

Drug: VIC-1911
VIC-1911 tablets for oral administration

Experimental: Expansion Phase, Cohort 2b: VIC-1911 plus sotorasib combination therapy
Subjects with locally advanced or metastatic KRAS G12C-mutated NSCLC refractory to or relapsed on prior KRASG12C inhibitor therapy will receive VIC-1911 plus sotorasib combination therapy.

Drug: VIC-1911
VIC-1911 tablets for oral administration

Drug: sotorasib
Sotorasib tablets for oral administration

Other Names:

LUMAKRAS

Experimental: Expansion Phase, Cohort 2c: VIC-1911 plus sotorasib combination therapy
Subjects with locally advanced or metastatic KRAS G12C-mutated NSCLC naive to KRAS G12C inhibitor therapy will receive VIC-1911 plus sotorasib combination therapy.

Drug: VIC-1911
VIC-1911 tablets for oral administration

Drug: sotorasib
Sotorasib tablets for oral administration

Other Names:

LUMAKRAS

What is the study measuring?

Primary Outcome Measures

Outcome Measure	Measure Description	Time Frame
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Incidence of Treatment Emergent Adverse Events (Safety and Tolerability)		
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Safety and tolerability assessed by adverse events(AEs) and serious adverse events (SAEs)		9 months
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Secondary Outcome Measures

Outcome Measure	Measure Description	Time Frame
Objective Response Rate	Proportion of subjects with objective responses (complete response [CR] + partial response [PR]) as assessed by Response Evaluation Criteria in Solid Tumors (RECIST) v.1.1.1. Assessed at the end of Cycle 2 and every 2 cycles thereafter up to 8 months. Each cycle is 28 days.	
Duration of Response	Length of time from first evidence of objective response (complete response [CR] or partial response [PR]) to the first objective evidence of disease progression. Assessed at the end of Cycle 2 and every 2 cycles thereafter up to 8 months. Each cycle is 28 days.	
Time to Response	Length of time from the date of first dose of study drug to the first evidence of objective response (CR,PR). Assessed at the end of Cycle 2 and every 2 cycles thereafter up to 8 months. Each cycle is 28 days.	
Disease Control Rate	Proportion of subjects with best response of CR, PR or stable disease (SD). Assessed at the end of Cycle 2 and every 2 cycles thereafter up to 8 months. Each cycle is 28 days.	
Progression-Free Survival	Length of time from the date of first dose of study drug to the first evidence of disease progression or death, whichever is earlier. Assessed from the date of the first dose of study drug to the first evidence of disease progression or death, whichever is earlier, assessed up to 9 months.	
Overall Survival	Length of time from the date of first dose of study drug to date of death from any cause. Assessed from the date of the first dose of study drug to date of death from any cause, assessed up to 9 months.	

Other Outcome Measures

Outcome Measure	Measure Description	Time Frame
Mean Plasma Concentrations of VIC-1911 Alone and in Combination With Sotorasib	Mean plasma concentrations of VIC-1911 will be determined and summarized by dose group. Was to be assessed C1D1; C1D15; C2D1; C4D1; C6D1 (each cycle is 28 days). The study terminated early with only 3 subjects analyzed C1D1 and 2 subjects analyzed C1D15.	
Circulating Tumor DNA (ctDNA) in Plasma (Pharmacodynamic Endpoint)	Changes from baseline will be determined, summarized by dose group and correlated with clinical outcome. Cycle 1 Day 1 pre-dose and at progression of disease.	
Tumor Biopsies for Biomarker Assessment (Pharmacodynamic Endpoint)	Changes from baseline will be determined, summarized by dose group and correlated with clinical outcome. Pre-study, Cycle 3 Day 1, and at progression of disease.	
Effect of de Novo Versus Acquired Resistance to KRASG12C Inhibitor Therapy, in Subjects Refractory to or Relapsed on Prior KRAS G12C Inhibitor Therapy	The effect of prior KRAS G12C de novo resistance (KRAS G12C inhibitor treatment ≤ 3 months) versus KRAS G12C acquired resistance (KRAS G12C inhibitor treatment > 3 months) on clinical outcome. 42 months.	

Collaborators and Investigators

This is where you will find people and organizations involved with this study.

Sponsor

Vitrac Therapeutics, LLC

Collaborators

Westat

Investigators

Study Chair: Sarah Goldberg, MD, MPH, Yale Cancer Center, Yale University

Study Director: Linda J Paradiso, DVM, Vitrac Therapeutics, LLC

Study Record Dates

These dates track the progress of study record and summary results submissions to ClinicalTrials.gov. Study records and reported results are reviewed by the National Library of Medicine (NLM) to make sure they meet specific quality control standards before being posted on the public website.

Study Registration Dates

First Submitted

2022-05-10

First Submitted that Met QC Criteria

2022-05-10

First Posted

2022-05-16

Results Reporting Dates
Results First Submitted
2024-05-14
Results First Submitted that Met QC Criteria
2025-04-29
Results First Posted
2025-05-01
Study Record Updates
Last Update Submitted that Met QC Criteria
2025-04-29
Last Update Posted
2025-05-01
Last Verified
2025-03